HTE LITE

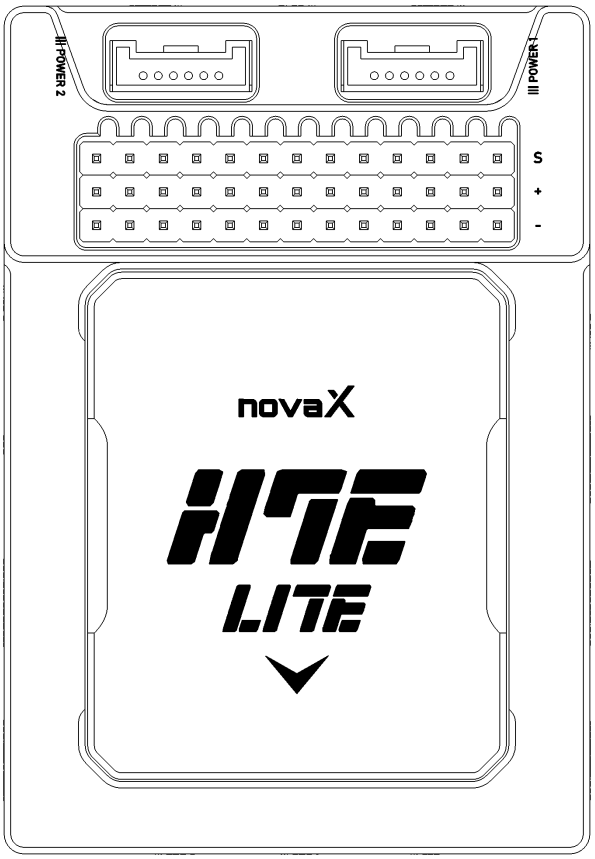
User Guide



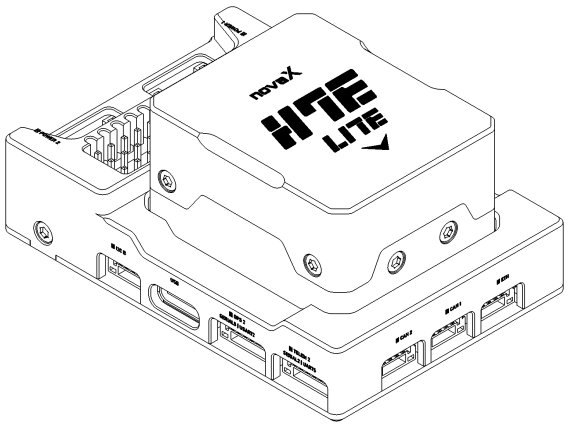
novaX

1. Parts List

· Main & Board



H7E Lite Autopilot



1. Parts List

· Connector Type

Port	Connector
POWER 1 · POWER 2	Molex Micro-Lock Plus 6P (1.25 mm)
TELEM 1 · TELEM 2 · GPS 1 · GPS 2	JST-GH 6P
SERIAL4 · SERIAL6 · I2C A · I2C B · CAN 1 · CAN 2 · ETHERNET	JST-GH 4P
RC IN	JST-GH 5P
DEBUG	JST-SH 6P
USB	USB Type-C
PWM OUT M1–M12 · SB (SBUS OUT)	2.54 mm 3 × 13 pin header

2. Specifications

· Hardware Specifications (1)

MCU	STM32H753, ARM Cortex-M7, 480 MHz
RAM / Flash	1 MB / 2 MB
IMU	2× ICM-42688-P + BMI088
IMU Heater	Built-in heater on the IMU board, held at 45 °C
Magnetometer	RM3100
Barometer	2× ICP-20100
Operating Voltage	4.75 – 5.7 V (Rated 5 V)
USB Input	4.75 – 5.25 V
Power Input	POWER 1 + POWER 2 (redundant, I2C power modules)
PWM Output	12 FMU channels (DShot except M7–M8) + SBUS out
RC Input	S.Bus, PPM, DSM / Spektrum
RSSI Input	Analog
UART	6 serial ports (TELEM 1 · 2, GPS 1 · 2, SERIAL4, SERIAL6) + debug console
GPS	GPS 1 + GPS 2 (6-pin, UART + I2C, left / right)

2. Specifications

· Hardware Specifications (2)

I²C	2 dedicated ports (I2C A, I2C B) + GPS 1 · 2 + power-module bus on POWER 1 · 2
CAN	2 Port
Ethernet	100 Mbps x 1 Port
Size	44.5 × 64.5 × 28.6 mm
Mount Hole	16 × 30 mm / M3 (4 blind holes in the base, 3 mm thread)
Supported F/W	novaX ArduPilot

2. Specifications

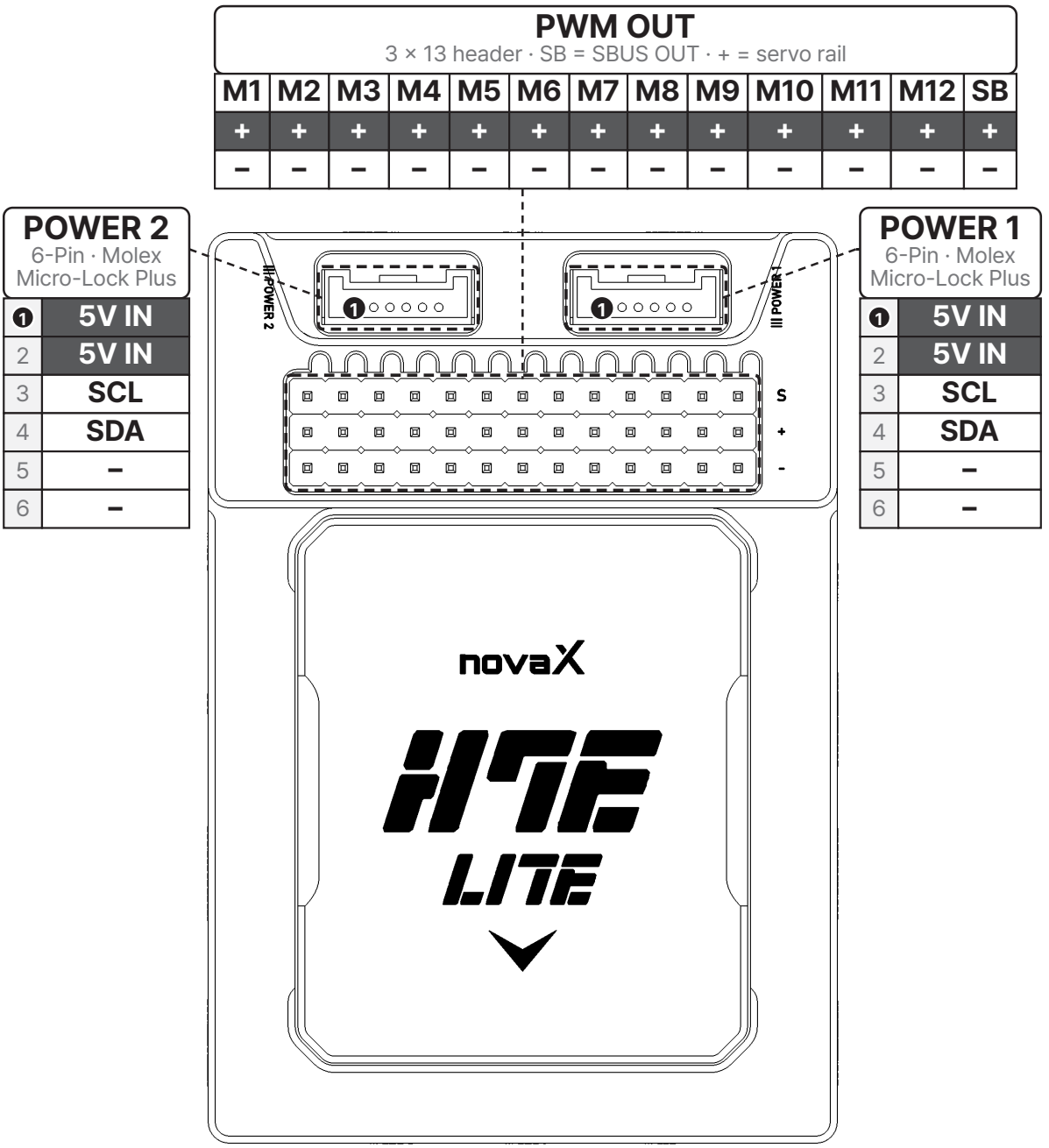
· Interfaces

POWER	2	ETHERNET	1
TELEM	2	DEBUG	1
GPS	2	USB	1
SERIAL	2	PWM Outputs	12
I2C	2	SBUS OUT	1
CAN	2	TF Card Slot	1
RC IN	1		
Power Module	POWER 1 + POWER 2 (redundant, I2C power modules)		

3. Pinouts

· Top

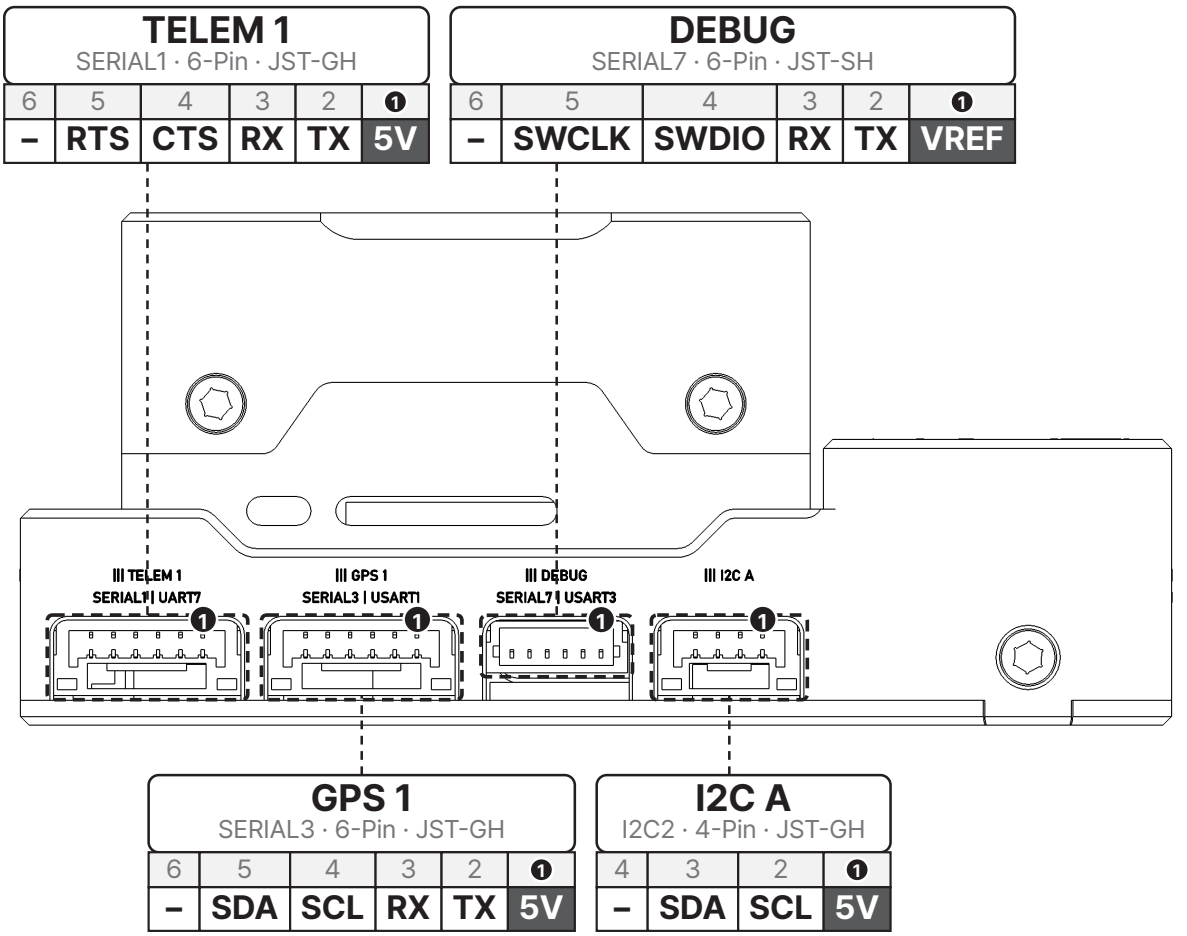
❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture



3. Pinouts

· Left

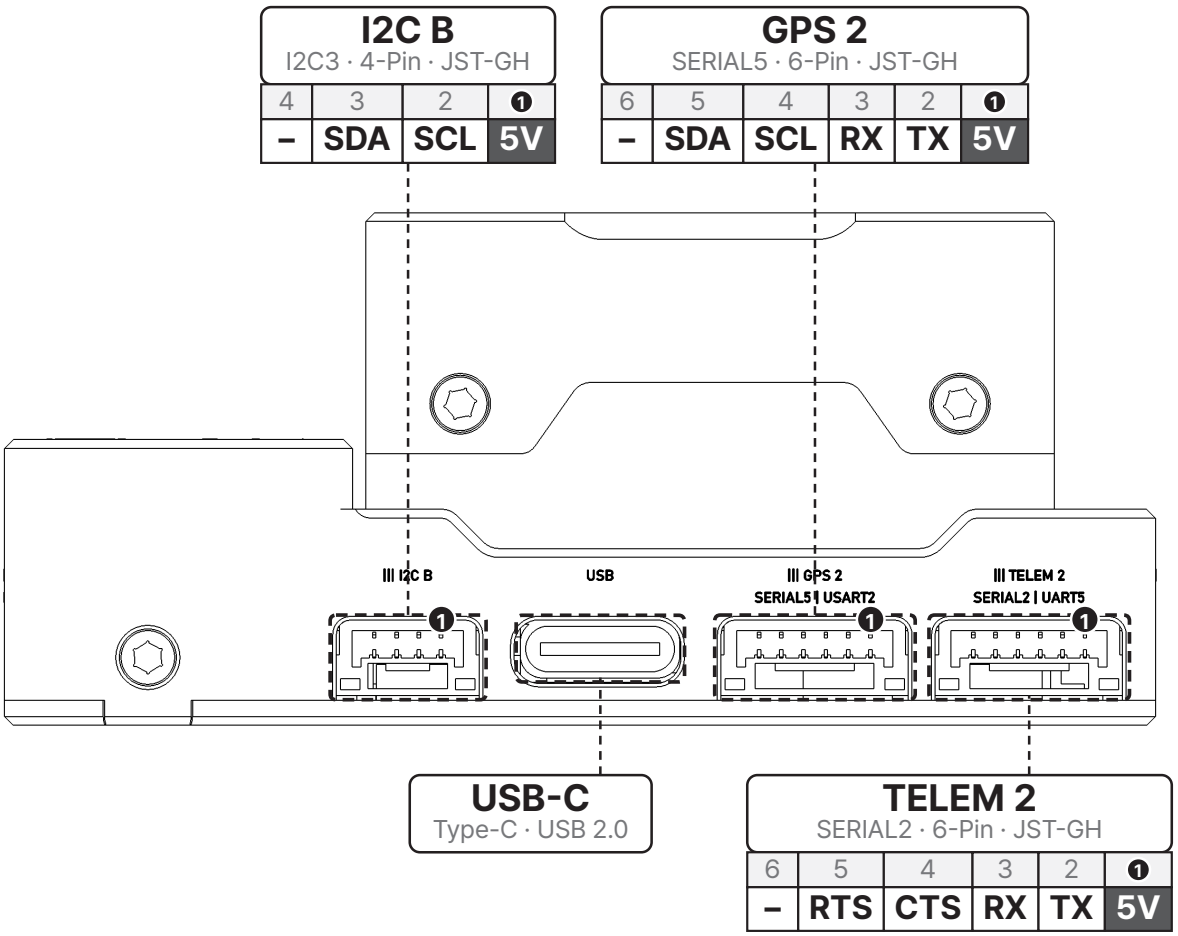
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3. Pinouts

· Right

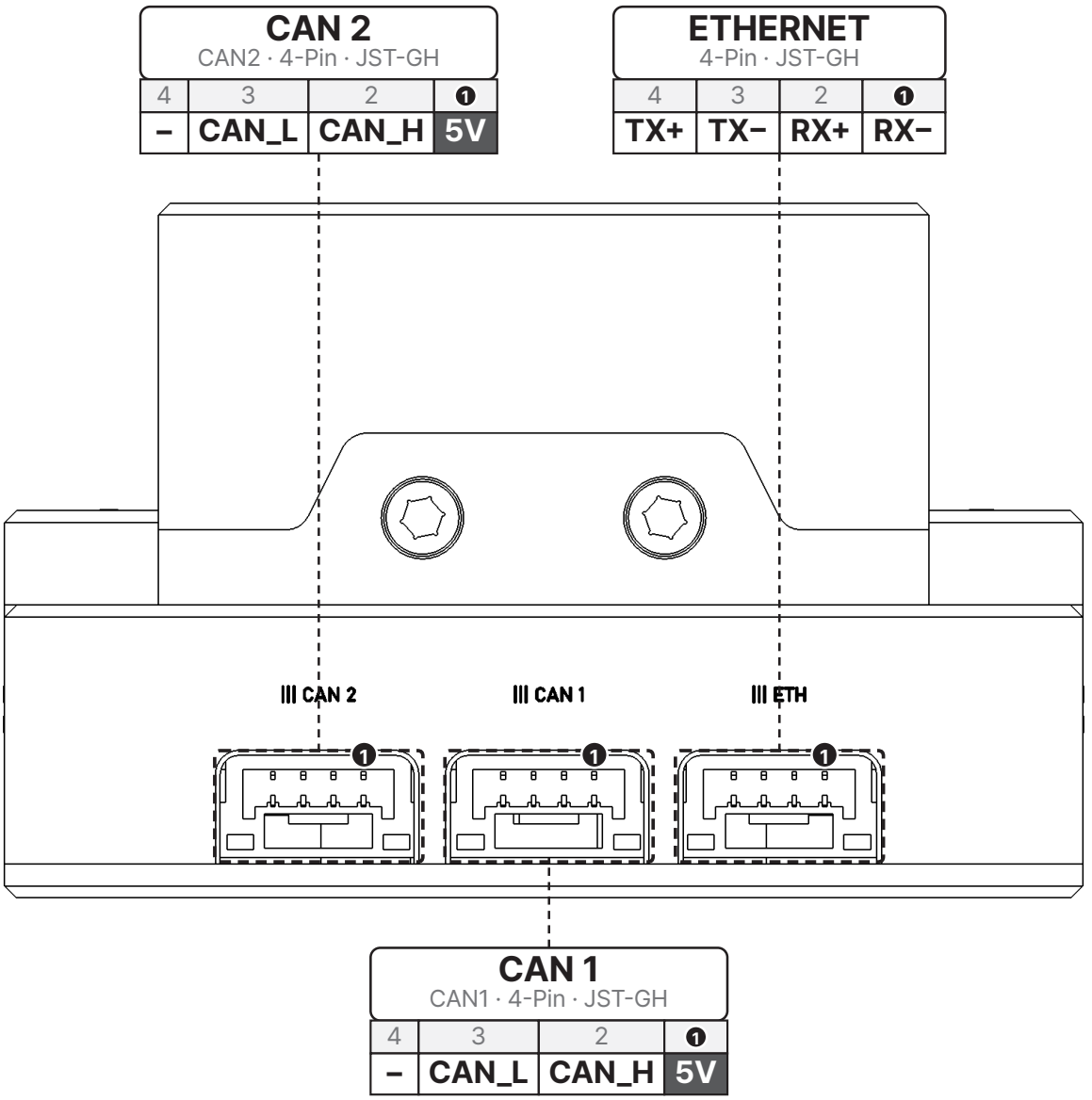
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3. Pinouts

· Front

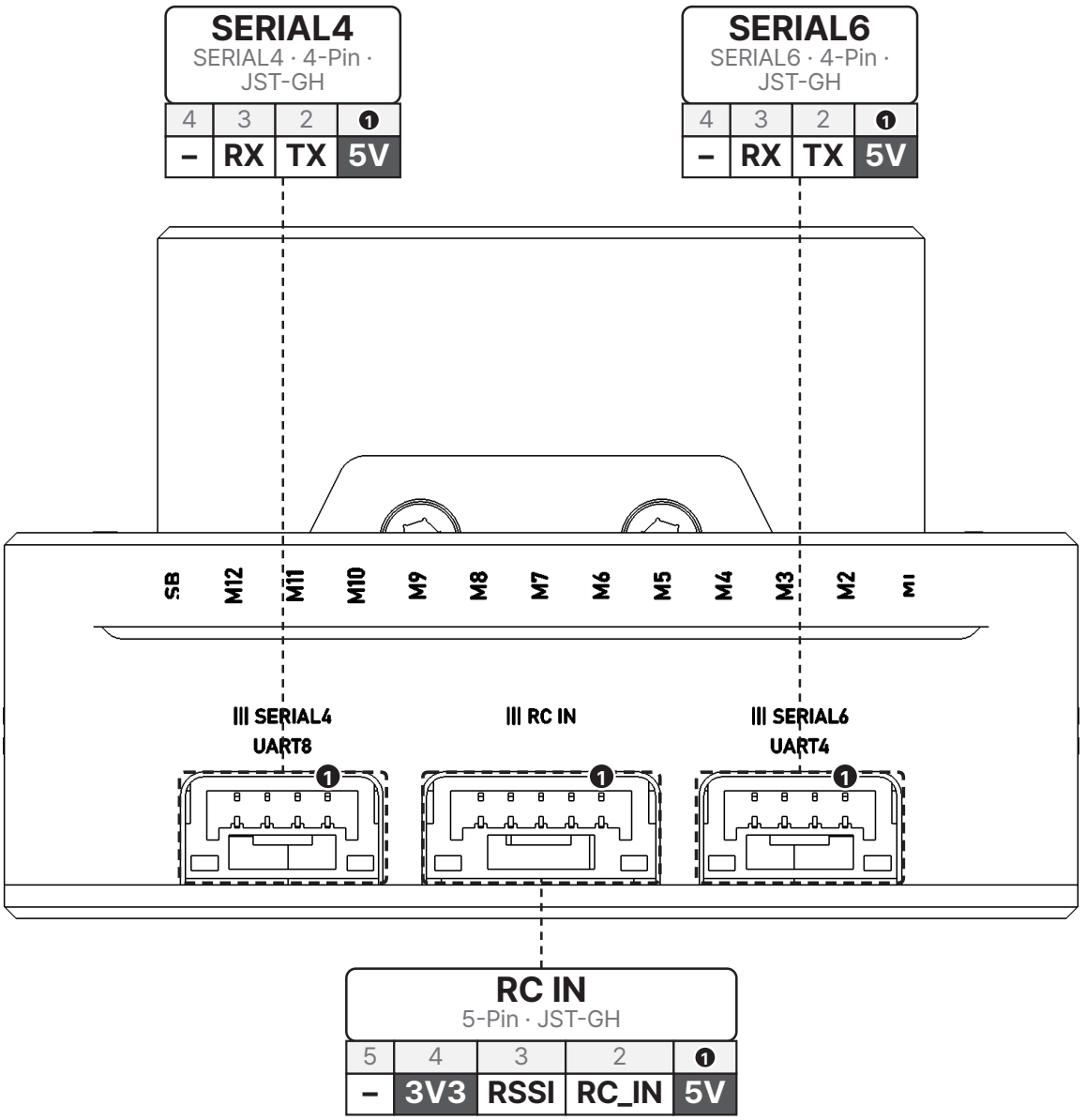
❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture



3. Pinouts

· Rear

❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture



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HTE LITE

3. Pinouts

· Top

❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture

PWM OUT
3 × 13 header · SB = SBUS OUT · + = servo rail

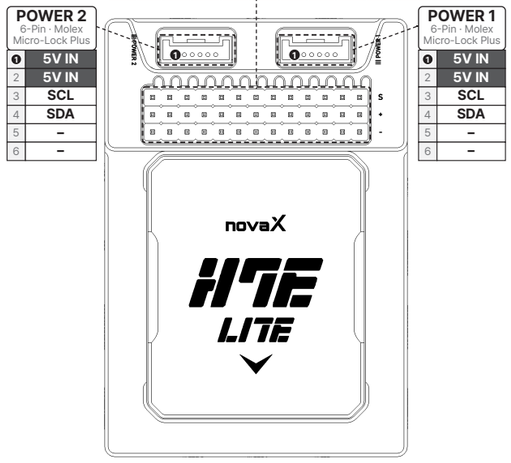
M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	SB
+	+	+	+	+	+	+	+	+	+	+	+	+
–	–	–	–	–	–	–	–	–	–	–	–	–

POWER 2
6-Pin · Molex Micro-Lock Plus

❶	5V IN
2	5V IN
3	SCL
4	SDA
5	–
6	–

POWER 1
6-Pin · Molex Micro-Lock Plus

❶	5V IN
2	5V IN
3	SCL
4	SDA
5	–
6	–



3. Pinouts

· Left

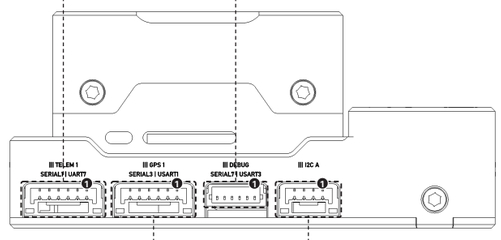
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TELEM 1
SERIAL1 · 6-Pin · JST-GH

6	5	4	3	2	❶
–	RTS	CTS	RX	TX	5V

DEBUG
SERIAL7 · 6-Pin · JST-SH

6	5	4	3	2	❶
–	SWCLK	SWDIO	RX	TX	VREF



GPS 1
SERIAL3 · 6-Pin · JST-GH

6	5	4	3	2	❶
–	SDA	SCL	RX	TX	5V

I2C A
I2C2 · 4-Pin · JST-GH

4	3	2	❶
–	SDA	SCL	5V

3. Pinouts

· Right

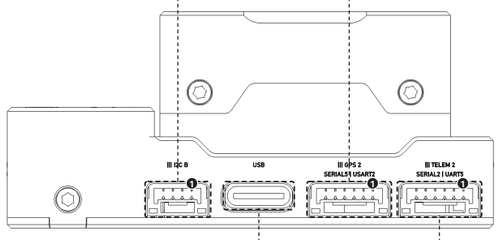
❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture

I2C B
I2C3 · 4-Pin · JST-GH

4	3	2	❶
–	SDA	SCL	5V

GPS 2
SERIAL5 · 6-Pin · JST-GH

6	5	4	3	2	❶
–	SDA	SCL	RX	TX	5V



USB-C
Type-C · USB 2.0

TELEM 2
SERIAL2 · 6-Pin · JST-GH

6	5	4	3	2	❶
–	RTS	CTS	RX	TX	5V

3. Pinouts

· Front

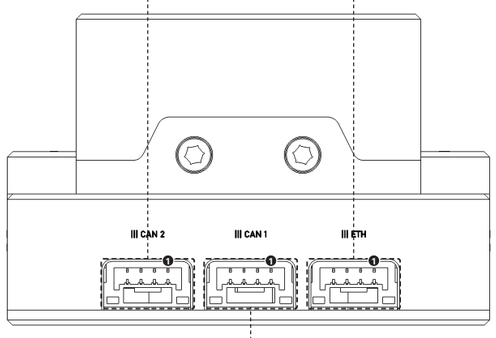
❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture

CAN 2
CAN2 · 4-Pin · JST-GH

4	3	2	❶
–	CAN_L	CAN_H	5V

ETHERNET
4-Pin · JST-GH

4	3	2	❶
TX+	TX–	RX+	RX–



CAN 1
CAN1 · 4-Pin · JST-GH

4	3	2	❶
–	CAN_L	CAN_H	5V

3. Pinouts

· Rear

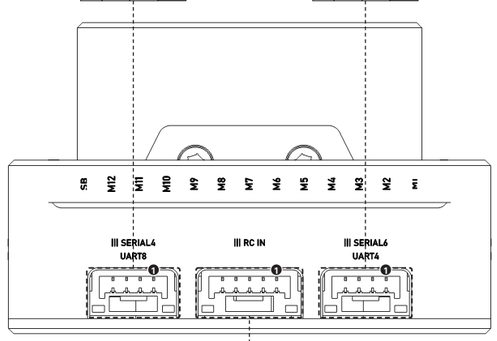
❶ = pin 1 **5V** = power **–** = GND **TX** = signal · pins as seen in the picture

SERIAL4
SERIAL4 · 4-Pin · JST-GH

4	3	2	❶
–	RX	TX	5V

SERIAL6
SERIAL6 · 4-Pin · JST-GH

4	3	2	❶
–	RX	TX	5V



RC IN
5-Pin · JST-GH

5	4	3	2	❶
–	3V3	RSSI	RC_IN	5V